

dios in the UK and US to develop this game, because the required pantomimes were not available in India. Besides, MoCap equipment is very expensive. We would rather hire studios for a few days, as even the voice recording for international games has to be done abroad."

Howzzat!

Sri Lankan spinner Muttiah Muralitharan's controversial 'doosra'—the leg spin delivery bowled with an off-spin action—was subjected to a series of tests at the University of Western Australia's human movement department, using twelve high-speed MoCap cameras shooting at a speed of 250 fps.

A team of bio mechanists studied this *doosra* under controlled conditions. Reflective markers were placed on Muralitharan's body, which were then captured by the cameras and computers. The actual analysis was computer-generated.

In 2003, the International Cricket Council (ICC) adopted an analytical software tool called e-Cricket Pro, produced by Bangalore-based Phoenix Global Solutions India (PGSI) Private Limited to train umpires for international games. This software was also used by the Indian cricket team for game training and analysis.

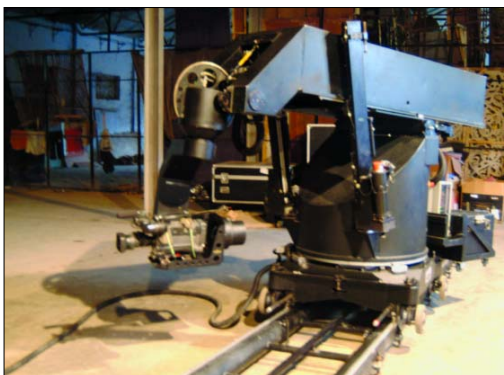
Many Indian players, including former Test cricketer Javagal Srinath and the inimitable Sachin Tendulkar—who studies his game on a laptop (one is provided to each member of the cricket team)—are believed to have benefited from motion analysis software. Motion analysis also aids injury prevention, and opportunities to study the effects of orthopaedic or athletic devices.

Apart from the Phoenix software, the Indian cricket team also uses a video-based motion analysis system called Silicon Coach, developed by Joe Morrison, a bio-mechanist from New Zealand. The system helps analyse players' techniques through four basic processes—error detection, analysing why the problem occurs, planning a solution to the problem, and finally, communicating with the player to implement



"Like MoCap software, Flair (used by Motion Control equipment) is compatible with most animation software such as Maya and 3D Studio Max,"

Sandeep Sethi
Manager- Motion Control,
Prime Focus Ltd



Motion control cameras can achieve super-human shots

the suggestions.

This software is marketed in India by S Ramakrishnan, a former First Division player, IBM certified computer professional and head of Meru Sports Technology Services, Chennai, which offers video-based motion analysis for the Indian cricket team. "Over time, small changes creep into a player's techniques without his or her knowledge. Video analysis helps you to identify these problems at an early stage, and possibly correct them," he says.

Silicon Coach is currently being used by the BCCI, National Cricket Academy, MRF Pace Foundation, Tamil Nadu Cricket Association, Mumbai Cricket Association, Bengal Cricket Association, TNTA and the Golf Federation. "We have also worked with world class coaches such as Dennis Lille, Bobby Simpson, Bom Woolmer, Peter Philport, Graham Gooch, Bruce Reid, John Wright and bio-mechanists from India and abroad, including Bruce Elliot and Mark Portus of the ICC," adds Ramakrishnan.

The 'Time Warp' feature in Silicon Coach provides instant feedback to players. A bowler can check and correct his action immediately after a delivery. "Though Silicon Coach is ideal

for self-analysis, the presence of a coach will definitely help the player," Ramakrishnan says.

TA Shekhar of the MRF Pace Foundation, one of the first institutions that invested in the software, says, "Almost all the boys who have trained at the Pace foundation have used Silicon Coach to analyse their style and performance. Bowlers like Irfan Pathan, Zaheer Khan and all

the others too."

Indian coach John Wright is also believed to be doing extensive research on opposition teams studying data compiled by computer analysts such as Ramakrishnan.



Hardware Requirements

Motion capture involves mapping 'data points' (usually at the joints) of the subject. The process involves the use of electromagnetic fields and magnetic sensors, or by using special markers that reflect the infra-red light that comes from illuminators, which is then captured by MoCap cameras.

The captured data is then analysed using specialised software (such as Kaydara MOCAP, for example), which renders the performer's movements on 3D characters in real-time. The data can then be edited and manipulated

using animation tools, or advanced mapping technologies by applying motion data to characters.

MoCap equipment is very expensive—the digital cameras, lights, and the software required

are extremely high-end, and very few studios invest in the equipment. The risk of obsolescence, too, runs high with any such technology.



The Vicon VCam MoCap camera